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Seminar paper

Betting Against Beta

A Study of Proposition 1 & 2 in American Equity Markets

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Abstract

This paper examines the beta market anomaly, where stocks with higher betas tend to have produce lower risk-adjusted returns than those with lower betas, contradicting a fundamental prediction by the CAPM-model. This is done by testing the proposition 1 and 2 from the paper of Frazzini and Pedersen (2014). Using the newest available data from US equity markets we test proposition 1 by examining the relationship between beta and alpha to test for the presence of the market anomaly. To test proposition 2, we construct a BAB-portfolio to examine the performance of a strategy which exploits this market anomaly by going long on low beta stocks and short on high beta stocks. We find evidence of a negative relationship between beta and alpha, which supports proposition 1 and that the beta anomaly is present leading to a flatter market security line. We also find that the BAB-portfolio generates positive returns, which supports proposition 2. However, when we introduce transaction costs into our analysis, we find that the performance of the BAB-portfolio is significantly reduced, suggesting that the strategy may not be as effective as initially thought when accounting for real-world trading frictions.

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Chapter 1

Introduction

One of the central predictions of the capital asset pricing model (CAPM) [22, 17] is a positive, linear relationship between expected returns and systematic risk. This relationship has long been challenged by empirical evidence suggesting the slope is far flatter than the model predicts. Early evidence of this was documented by Haugen and Heins [14], who showed that low-risk portfolios historically earned higher returns than high-risk portfolios. This is known as the beta anomaly, whereby high-beta assets yield lower risk-adjusted returns than CAPM predicts, directly contradicting the model, as documented by Fama and French [10] among others. One way of exploiting this anomaly is the investing strategy of Betting-Against-Beta (BAB).

Introduced by Frazzini and Pedersen (2014) [13], the BAB strategy is a long-short strategy which exploits the beta anomaly by going long on low-beta stocks and short on high-beta stocks. The portfolio is formed by cross-sectional sorting on beta, and the strategy has been shown to generate positive risk-adjusted returns (alpha) across different markets and asset classes [13]. However, its performance can be questioned, with some studies suggesting that the returns may be driven by other factors such as size and value effects [9], or that the strategy is not viable after accounting for transaction costs [19].

To examine the relevance of the beta anomaly we test Propositions 1 and 2 from Frazzini and Pedersen (2014) [13]. The first proposition states that there is a negative relationship between beta and alpha, implying that stocks with higher betas tend to have lower alphas. The second proposition states that a BAB portfolio exploiting this anomaly should generate positive risk-adjusted returns (alpha).

For clarity, the paper is structured as follows: first, we introduce the theoretical background for the CAPM and why leverage constraints generate the beta anomaly. Second, we describe our data and methodology for testing the two propositions and how we extend the analysis by introducing transaction costs. Third, we present the results. Fourth, we connect our results to the existing literature. Lastly, we conclude and suggest avenues for future research. The goal of the structure is to ensure replicatbility and transparency in the analysis for further reseach.

Using US equity data for the period 1963–2025, we find a negative relationship between beta and alpha supporting Proposition 1, with the empirical Security Market Line significantly flatter than CAPM predicts. We also find that the BAB portfolio generates positive risk-adjusted returns, supporting Proposition 2. However, when we introduce transaction costs, the performance of the BAB portfolio is significantly reduced, suggesting that the strategy may not be viable after accounting for real-world trading frictions.

Chapter 2

Theory

In this chapter we discuss the theoretical background needed to understand the results and methodology. This entails firstly describing the CAPM as the baseline model of comparison and its predictions. Secondly, we discuss the shortcomings of the CAPM and how the presence of leverage constraints can lead to a flatter security market line thus creating the beta anomaly as discussed by Frazzini and Pedersen (2014) [13]. Lastly we introduce Propositions 1 and 2 from the same paper which we wish to test.

2.1 The Capital Asset Pricing Model

Our baseline model of interest is the capital asset pricing model (CAPM), which seeks to explain the relationship between expected returns and market risk, the component of risk that cannot be diversified away. The model predicts this relationship to be linear and positive, with the expected return of an asset equal to the risk-free rate plus a risk premium proportional to the asset's beta, which measures the sensitivity of the asset's returns to the market portfolio. The CAPM makes several underlying assumptions. These include: a single-period economy in equilibrium, no transaction costs or taxes, universally available risk-free borrowing and lending, homogeneous investor expectations, and mean-variance optimising investors. [22, 17] The model can be specified as follows:

$$R_i - r_f = \alpha + \beta(R_m - r_f) + \epsilon \tag{2.1}$$

Where $R_i - r_f$ is the excess return of asset i, r_f is the risk-free rate, α is the intercept, β is the slope coefficient which measures the sensitivity of the asset's returns to the returns of the market portfolio, $R_m - r_f$ is the excess return of the market portfolio and ϵ is the error term. The CAPM predicts that in equilibrium, all assets should lie on the security market line (SML) which is a graphical representation of the relationship between expected returns and beta. The SML is upward sloping, indicating that higher beta assets should have higher expected returns.

Empirical evidence, however, has long challenged the predictions of the CAPM. One of the most well-known anomalies is the beta anomaly, which refers to the empirical finding that stocks with higher betas tend to have lower alphas than those with lower betas. This contradicts the predictions of the CAPM and suggests that there may be other factors at play in determining expected returns. The paper of focus here is Frazzini and Pedersen (2014) [13], which seeks to explain the beta anomaly and test the propositions that arise from their theoretical framework.

2.2 Leverage Constraints and a Flatter Security Market Line

The paper of Frazzini and Pedersen (2014)[13] suggests that the presence of leverage constraints can lead to a flatter security market line, which can explain the beta anomaly. Leverage constraints refer to the limitations that investors face when trying to borrow money to invest in higher beta assets. These constraints can arise from various sources, such as margin requirements, borrowing limits, or risk management policies. Investors who want leveraged risk exposure but face borrowing constraints achieve their target risk by tilting toward high-beta assets instead of leveraging low-beta ones. This creates excess demand for high-beta assets, bidding up their prices and lowering their expected returns and alphas, producing the flatter security market line that characterises the beta anomaly.

A precursor to this mechanism is found in Black [5], whose zero-beta CAPM model shows that even without short-selling constraints a flat SML can arise when investors cannot borrow at the risk-free rate. In Black's model, investors who face borrowing restrictions hold the market portfolio levered via risky assets, bidding up high-beta stocks and depressing their expected returns, the same prediction as Frazzini and Pedersen, derived from a different friction. Asness, Frazzini, and Pedersen [3] further develop this idea under the label of leverage aversion, showing that investors who are averse to or constrained from using leverage systematically overweight high-risk assets, which compresses the risk-return tradeoff and generates a flat SML as a structural feature of equilibrium.

A key reason the anomaly persists even after becoming widely documented is that arbitrageurs face their own funding constraints. Shleifer and Vishny [23] show that arbitrage capital is limited and that arbitrageurs who bet against overpriced assets may be forced to unwind positions at a loss if prices move further against them before converging. This limits-to-arbitrage mechanism is directly relevant since the same funding constraints that create the beta anomaly also prevent it from being fully exploited and eliminated.

2.3 Proposition 1: High Beta is Low Alpha

Frazzini and Pedersen (2014)[13] show that in the presence of funding constraints, the equilibrium expected return for any security s is:

$$E_t(r^s_{t+1}) = r^f + \psi_t + \beta^s_t \lambda_t \tag{2.2}$$

where $\lambda_t = E_t(r^M_{t+1}) - r^f - \psi_t$ is the risk premium adjusted for funding constraints and ψ_t is the average Lagrange multiplier that measures the tightness of the funding constraints. A security's alpha with respect to the market is then:

$$\alpha^s_t = \psi_t(1 - \beta^s_t) \tag{2.3}$$

The alpha decreases in beta β^s_t. When $\beta^s_t > 1$, alpha is negative, high-beta stocks are overpriced. When $\beta^s_t < 1$, alpha is positive, low-beta stocks are underpriced. Furthermore, for an efficient portfolio, the Sharpe ratio is highest for a beta below one, decreasing in β^s_t for higher betas and increasing for lower betas.

2.4 Proposition 2: The BAB Factor

To construct the BAB factor, let w^L_t be the relative portfolio weights for a portfolio of low-beta assets with return $r^L_{t+1} = w^L_t r_{t+1}$, and similarly a portfolio of high-beta assets with return r^H_{t+1}. The betas of these portfolios are denoted β^L_t and β^H_t, where $\beta^L_t < \beta^H_t$. The BAB factor return is then:

$$r^{BAB}_{t+1} = \frac{1}{\beta^L_t}(r^L_{t+1} - r^f) - \frac{1}{\beta^H_t}(r^H_{t+1} - r^f) \tag{2.4}$$

This portfolio is market-neutral with a beta of zero. The long side is leveraged to a beta of one and the short side is de-leveraged to a beta of one. Frazzini and Pedersen (2014) [13] show that the expected excess return of this self-financing BAB factor is positive:

$$\boxed{E_t(r^{BAB}_{t+1}) = \frac{\beta^H_t - \beta^L_t}{\beta^L_t \beta^H_t}\psi_t \geq 0} \tag{2.5}$$

The expected return is increasing in the ex ante beta spread $(\beta^H_t - \beta^L_t)/(\beta^L_t \beta^H_t)$ and funding tightness ψ_t. Thus a market-neutral BAB portfolio that is long leveraged low-beta securities and short higher-beta securities earns a positive expected return on average, with the size depending on how binding the funding constraints are in the market.

2.5 Market Neutrality and Self-Financing

The BAB factor has two important portfolio properties that are worth making explicit, namely that it is both market-neutral and self-financing. These are distinct properties that hold independently and simultaneously.

Market neutrality means the portfolio has zero beta with respect to the market, so its returns are unaffected by broad market movements. Frazzini and Pedersen [13] explicitly construct the BAB factor with this property by scaling each leg by the inverse of its beta, as in Equation 2.4. The long leg, with raw portfolio beta β_t^L, is scaled by $1/\beta_t^L$, giving it an effective market beta of exactly one. Similarly, the short leg with raw beta β_t^H is scaled by $1/\beta_t^H$, also giving it an effective beta of one. Since the strategy is long one unit of market beta and short one unit of market beta, the combined portfolio beta is:

$$\beta^{BAB} = \frac{1}{\beta_t^L} \cdot \beta_t^L - \frac{1}{\beta_t^H} \cdot \beta_t^H = 1 - 1 = 0$$

Importantly this ensures that any return earned by the BAB factor reflects the pricing difference between low- and high-beta stocks, the anomaly itself, rather than passive exposure to market returns. Without this scaling, a strategy that simply bought low-beta and shorted high-beta stocks would have non-zero net market exposure and its returns would be contaminated by market movements.

Self-financing means the strategy requires no net external capital, a property Frazzini and Pedersen [13] build into the BAB factor by construction. The short position in high-beta stocks generates proceeds which are used to fund the leveraged long position in low-beta stocks. The net cash outflow at inception is therefore zero. This property makes the BAB factor directly interpretable as a zero-cost trading strategy, and its return represents pure profit per unit of capital employed in the leveraging and de-leveraging of the two legs. In practice, however, an investor would need to post margin for the short position, meaning some capital is tied up as collateral — consistent with the funding constraint mechanism that motivates the beta anomaly in the first place.

2.6 Transaction Costs and the Bid-Ask Spread

A central question in the factor investing literature is whether anomaly returns survive once realistic trading frictions are accounted for. Transaction costs can be decomposed into several components: explicit costs such as commissions and taxes, and implicit costs such as the bid-ask spread and market impact. For liquid, exchange-traded equities, the bid-ask spread is generally considered the most important implicit cost and the most directly observable measure of the cost of immediacy [20].

The bid-ask spread arises because market makers face adverse selection risk from informed traders and must be compensated for providing liquidity [15]. The quoted spread, the difference between the ask price at which an investor buys and the bid price at which they sell, represents the round-trip cost of a trade executed at the prevailing quotes. An alternative measure of illiquidity is the Amihud ratio [1], which captures price impact as the ratio of absolute return to trading volume, but this measure is better suited to measuring the cost of large trades rather than the per-share cost of routine rebalancing.

For the purpose of evaluating the BAB strategy, the bid-ask spread is the natural cost measure because the strategy involves frequent rebalancing across a broad universe of stocks, with each trade being a small fraction of daily volume. In this setting, market impact is likely to be modest and the spread dominates the cost of execution. This approach is consistent with Frazzini, Israel, and Moskowitz [12], who use bid-ask spreads as a baseline cost measure before incorporating actual execution data from a large institutional investor. They find that effective spreads, the costs actually incurred by a large asset manager trading inside the quoted spread — are substantially lower than quoted spreads, implying that quoted spread-based estimates provide an upper bound on true transaction costs.

An alternative approach to estimating transaction costs that does not require bid-ask data is the zero-return estimator of Lesmond et al. [16], which infers round-trip costs from the frequency of zero-return days on the assumption that informed traders only trade when the signal exceeds the transaction cost. While this estimator is better suited to less liquid markets where bid-ask data may be unavailable, it provides a useful benchmark for validating spread-based estimates. More broadly, Novy-Marx and Velikov [19] document that transaction costs are a first-order concern for most published anomalies, with many strategies becoming unprofitable after realistic cost estimates.

Chapter 3

Data and Methodology

In this chapter we describe the data and methodology used to test the propositions of Frazzini and Pedersen (2014) [13]. We first summarize the characteristics of the data and their sources, then we detail the beta estimation procedure, the construction of the BAB factor, and the Fama-MacBeth regressions used to test Propositions 1 and 2. Finally, we discuss how we introduce and estimate transaction costs for the BAB strategy, with code included for transparency and reproducibility.

3.1 Data Description

This paper replicates the Betting Against Beta (BAB) factor of Frazzini and Pedersen [13] for the US equity market using CRSP data accessed via Wharton Research Data Services (WRDS) through the `tidyfinance` package [21]. The sample covers January 1963 through December 2025. The choice of this period is driven by data availability and to ensure a long enough time series for robust beta estimation and portfolio construction.

Table 3.1: Data sources used in the analysis. CRSP data is accessed via WRDS through the `tidyfinance` Python package. Fama-French factors are downloaded from the Kenneth French data library. All data covers January 1963 to December 2025.

Dataset	Frequency	Content	Purpose
CRSP stock data	Monthly	Excess returns, market capitalisation (NYSE, AMEX, NASDAQ)	BAB portfolio construction and factor returns
CRSP stock data	Daily	Excess returns, bid-ask prices	Volatility component of beta; proportional half-spreads for transaction costs
Fama-French factors	Monthly & daily	Risk-free rate, market excess return	Excess return computation; correlation component of beta

3.2 Fama-French Factors

We download Fama-French 3-factor data at monthly and daily frequency. We retain only the risk-free rate $r_{f,t}$ and market excess return, which are the inputs needed for excess return computation and beta estimation. The excess return for stock i in month t is defined as:

$$r_{i,t}^{excess} = r_{i,t} - r_{f,t}$$

3.3 Monthly CRSP Data

We download monthly CRSP data for all US common stocks listed on NYSE, AMEX, and NASDAQ. Market capitalisation is computed as shares outstanding times the absolute value of price, expressed in millions of USD:

$$MC_{i,t} = \frac{\text{shrout}_{i,t} \times |P_{i,t}|}{1{,}000}$$

The absolute value is required because CRSP records negative prices to indicate bid-ask midpoints rather than transaction prices. We also construct a one-month lagged market cap, which serves as the portfolio weight in the BAB factor construction.

3.4 Data Cleaning

Observations with missing returns or market capitalisation are excluded from the sample. To remove data errors in CRSP raw returns, excess returns are winsorised at -100%:

$$r_{i,t}^{excess} = \max(r_{i,t} - r_{f,t},\ -1)$$

Lagged market capitalisation (one month) is used as portfolio weights to ensure all weighting variables are known at the time of portfolio formation.

3.5 Daily CRSP Data

Beta estimation in Frazzini and Pedersen (2014)[13] requires daily returns. The volatility component of beta uses a 1-year rolling window (252 trading days, minimum 120 observations) of daily excess returns. We also retain bid and ask prices from CRSP to compute proportional half-spreads for the transaction cost analysis. We download the daily data in batches of 500 PERMNOs to manage memory.

3.6 Sample Description

Table 3.6 reports summary statistics, Figure 3.1 shows stock coverage over time, and Figure 3.2 shows the cross-sectional distribution of estimated betas.

Table 3.6: *Summary statistics for the final analysis sample covering January 1963 to December 2025. Excess Return is monthly and expressed as a decimal. Market Cap is in millions of USD. Beta is estimated following Frazzini and Pedersen (2014) as the product of a 60-month rolling correlation and a 252-day rolling volatility ratio, shrunk toward one. Source: CRSP via WRDS and Kenneth French data library.*

	Mean	Std	P5	P25	Median	P75	P95
Excess Return	0.0094	0.1779	-0.2132	-0.0666	-0.0029	0.0679	0.2508
Beta	1.0005	0.3877	0.4913	0.7252	0.9514	1.2100	1.6893
Mkt Cap ($M)	3166.3312	31325.4948	4.1318	26.6612	128.3700	758.7904	9328.1884

Table Table 3.6 shows that the sample contains 2,576,149 stock-month observations across 20,122 unique stocks. The mean monthly excess return is 0.94%, consistent with the well-documented equity risk premium, though the distribution is wide with a standard deviation of 17.79% per month. The mean estimated beta is 1.00, slightly above one, with a standard deviation of 0.39, confirming meaningful cross-sectional dispersion in market risk — a necessary condition for the BAB strategy to be viable. Market capitalisation is highly right-skewed, reflecting the dominance of a small number of large-cap stocks in the US equity market.

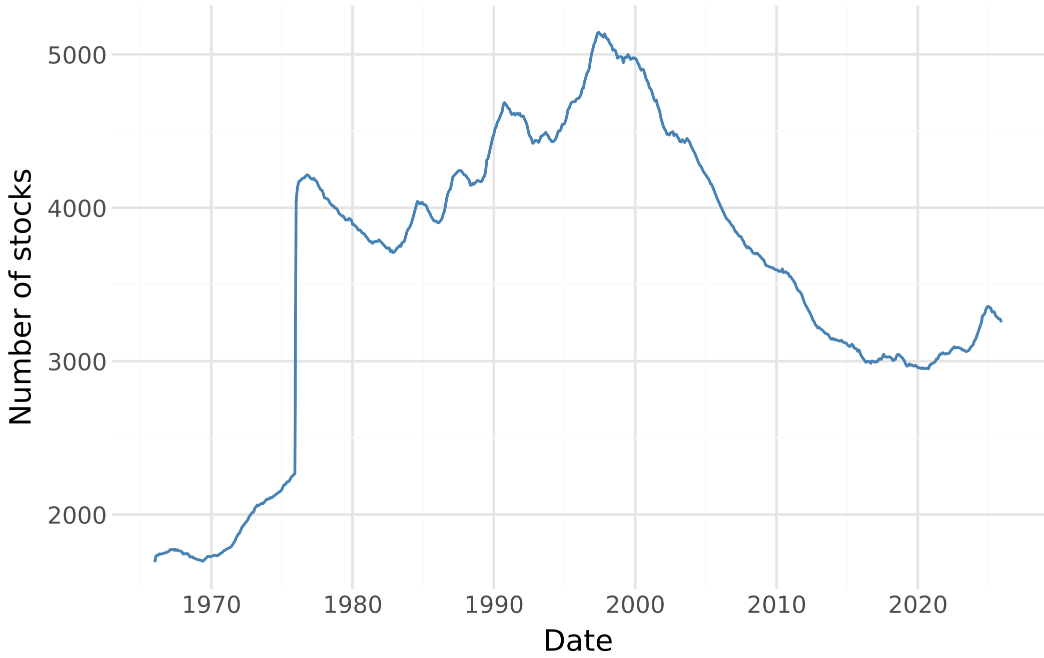

Figure 3.1: Number of distinct stocks included in the sample each month. Coverage grows steadily from around 500 stocks in the early 1960s to a peak of over 4,000 in the late 1990s before declining as NASDAQ listings contracted following the dot-com bubble. The pattern reflects the expansion of US equity markets and the availability of valid beta estimates.

Figure Figure 3.1 shows that coverage grows steadily from around 500 stocks in the early 1960s to a peak of over 4,000 in the late 1990s, before declining as NASDAQ listings contracted following the dot-com bubble. The variation in coverage is relevant because the BAB factor is constructed from the full cross-section each month — a larger and more diverse universe of stocks in the later part of the sample provides more dispersion in beta and therefore a more powerful test of the propositions.

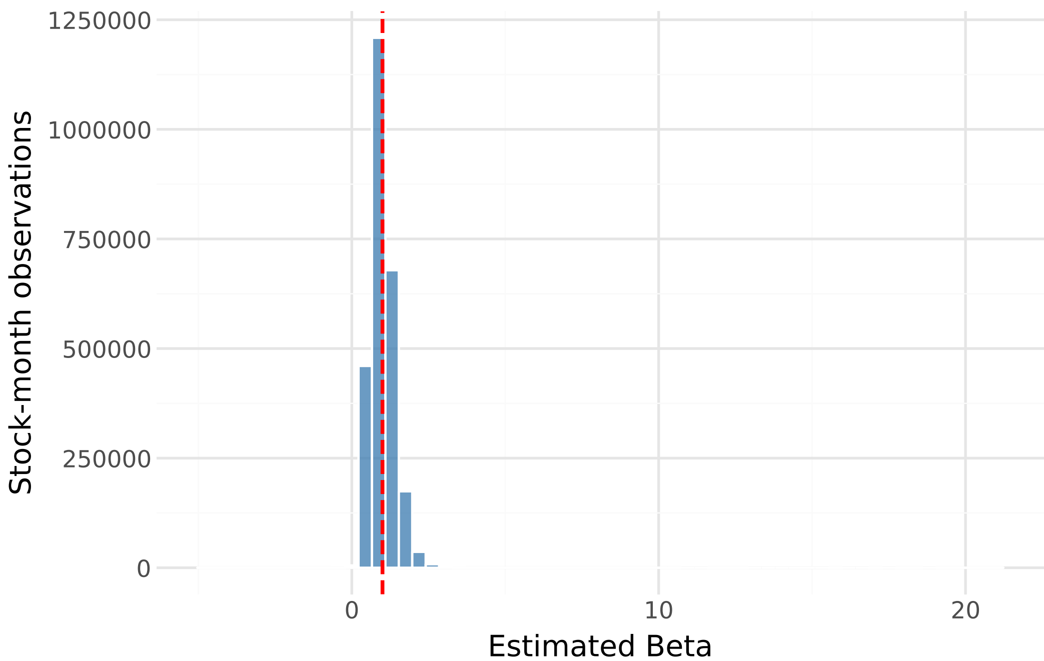

Figure 3.2: Cross-sectional distribution of estimated betas across all stock-month observations in the sample. The distribution is centred slightly above one with a long right tail driven by high-volatility small-cap stocks. The red dashed line marks a beta of one, which corresponds to the market portfolio under the CAPM.

Figure Figure 3.2 shows that the cross-sectional beta distribution is centred slightly above one and right-skewed, with a long tail driven by high-volatility small-cap stocks. The bulk of the distribution lies between 0 and 2, with most stocks having a beta below one — meaning the low-beta portfolio contains more stocks than a simple median split might suggest. The right tail confirms that there is a meaningful subset of high-beta stocks to short, providing the asymmetry in risk exposure that the BAB strategy exploits. The Vasicek shrinkage toward one compresses extreme estimates, which reduces the influence of the right tail on portfolio construction.

3.7 Beta Estimation

Following Frazzini and Pedersen [13], we estimate beta as the product of a correlation component and a volatility component:

$$\hat{\beta}_i = \hat{\rho}_i \cdot \frac{\hat{\sigma}_i}{\hat{\sigma}_m}$$

The correlation $\hat{\rho}_i$ is estimated from overlapping three-day log returns on a 5-year (1,260 trading day) rolling window with a minimum of 750 days, following the non-synchronous trading correction of Frazzini and Pedersen [13]. The volatility ratio is estimated from a 1-year rolling window of daily excess returns (minimum 120 observations). This two-component approach improves precision by using higher-frequency data for the noisier volatility estimate, while the overlapping three-day construction for the correlation mitigates non-synchronous trading bias for infrequently traded stocks.

3.7.1 Volatility Component

We estimate annualised standard deviations using a 252-day rolling window of daily excess returns. For stock i on day d, using a window of $T = 252$ days:

$$\hat{\sigma}_{i,d} = \sqrt{\frac{1}{T-1} \sum_{\tau=d-T+1}^{d} \left(r_{i,\tau}^{excess} - \bar{r}_{i,d}^{excess} \right)^2}$$

where $\bar{r}_{i,d}^{excess}$ is the mean excess return over the window. The market standard deviation $\hat{\sigma}_{m,d}$ is estimated identically. Both are estimated over the same window to ensure comparability in the volatility ratio $\hat{\sigma}_i/\hat{\sigma}_m$. For each stock-month, we retain the last available daily estimate within the month.

3.7.2 Correlation Component

We estimate the correlation with the market $\hat{\rho}_i$ using overlapping three-day log returns, following Frazzini and Pedersen [13]:

$$r_{i,t}^{3d} = \sum_{k=0}^{2} \ln(1 + r_{i,t+k})$$

The three-day aggregation corrects for non-synchronous trading: if a stock does not trade on day t, its observed return is zero while the market moves, causing the daily correlation to be underestimated. Summing over three consecutive days ensures that the delayed price discovery of infrequently traded stocks is captured. Overlapping three-day returns are computed for both the individual stock and the market from daily CRSP and Fama-French data. The rolling correlation uses a 1,260-trading-day (five-year) window with a minimum of 750 observations (approximately three years), and the last daily estimate within each calendar month is retained as the monthly correlation estimate.

3.7.3 Combined Beta Estimate

The two components are combined into a time-series beta $\hat{\beta}_i^{ts} = \hat{\rho}_i \cdot (\hat{\sigma}_i / \hat{\sigma}_m)$, which is then shrunk toward the cross-sectional mean of one using Vasicek shrinkage [24] with weights $(0.6, 0.4)$ as in Frazzini and Pedersen [13]:

$$\hat{\beta}_i = 0.6 \cdot \hat{\beta}_i^{ts} + 0.4 \cdot 1$$

Beta estimates are lagged by one month before merging with returns to avoid a look-ahead bias in portfolio construction.

3.8 Proposition 1: The Security Market Line

Proposition 1 predicts that the empirical SML is flatter than CAPM implies: high-beta assets earn lower risk-adjusted returns (negative CAPM alpha) and low-beta assets earn higher risk-adjusted returns (positive CAPM alpha). We test this by running Fama-MacBeth cross-sectional regressions of excess returns on lagged betas. In the first pass, a separate OLS regression is estimated for each month t:

$$r_{i,t}^{excess} = \gamma_{0,t} + \gamma_{1,t} \hat{\beta}_{i,t-1} + \epsilon_{i,t}$$

yielding a time series of monthly slope estimates $\{\hat{\gamma}_{1,t}\}$ and intercept estimates $\{\hat{\gamma}_{0,t}\}$. In the second pass, the reported coefficients are the time-series averages $\bar{\gamma}_j = T^{-1} \sum_t \hat{\gamma}_{j,t}$, and the associated t-statistics use Newey-West standard errors with four lags [18] to account for potential autocorrelation in the monthly coefficient estimates, following Frazzini and Pedersen [13]. Under the CAPM, $\gamma_{1,t}$ should equal the market excess return and $\gamma_{0,t}$ should equal zero; the beta anomaly predicts $\bar{\gamma}_1$ will be significantly below the average market excess return.

3.8.1 Beta-Sorted Portfolio Alphas

We sort stocks into 10 beta-decile portfolios and compute each decile's CAPM alpha to visualise the alpha-beta relationship. Within each decile, returns are value-weighted using lagged market capitalisation as weights. For each decile k, we estimate the time-series regression:

$$r_t^{k,excess} = \alpha_k + \beta_k r_t^{mkt,excess} + \epsilon_{k,t}$$

where α_k is the CAPM alpha for decile k. Under the beta anomaly, α_k should decrease monotonically from the lowest to the highest beta decile.

3.9 Proposition 2: The BAB Factor

Following Frazzini and Pedersen [13], at each month t we rank all stocks by their estimated beta across the full cross-section. Let $z_i = \text{rank}(\hat{\beta}_i)$ and $\bar{z} = \frac{1}{n} \sum_i z_i$ be the average rank. The low-beta portfolio contains all stocks with $z_i < \bar{z}$ and the high-beta portfolio all stocks with $z_i > \bar{z}$. Portfolio weights are:

$$w_i^L = k(\bar{z} - z_i), \quad w_i^H = k(z_i - \bar{z}), \quad k = \frac{2}{\sum_i |z_i - \bar{z}|}$$

where k is a single normalizing constant applied to both legs, ensuring each portfolio's weights sum to one and that stocks furthest from the median in beta space receive the largest weight. The BAB factor return is:

$$r_t^{BAB} = \frac{1}{\beta_t^L}(r_t^L - r^f) - \frac{1}{\beta_t^H}(r_t^H - r^f)$$

where $\beta_t^L = \sum_i w_i^L \hat{\beta}_i$ and $\beta_t^H = \sum_i w_i^H \hat{\beta}_i$ are the rank-weighted average betas of the two portfolios.

We evaluate the performance of the BAB factor using four summary statistics. The annualised mean return is $\bar{r}^{BAB} \times 12$ and the annualised standard deviation is $\hat{\sigma}^{BAB} \times \sqrt{12}$, where $\bar{r}^{BAB}$ and $\hat{\sigma}^{BAB}$ are the sample mean and standard deviation of the monthly BAB returns. The Sharpe ratio is the ratio of annualised mean excess return to annualised standard deviation:

$$SR = \frac{\bar{r}^{BAB} \times 12}{\hat{\sigma}^{BAB} \times \sqrt{12}} = \frac{\bar{r}^{BAB}}{\hat{\sigma}^{BAB}} \times \sqrt{12}$$

Since the BAB factor is already expressed as an excess return (it is self-financing and earns returns above the risk-free rate by construction), no additional subtraction of the risk-free rate is required. The t-statistic tests whether the mean monthly return is significantly different from zero.

3.10 Transaction Costs

We estimate the round-trip transaction cost for the BAB strategy using daily bid-ask spreads from CRSP. The proportional half-spread on day d for stock i is

$$c_{i,d} = \frac{\text{ask}_{i,d} - \text{bid}_{i,d}}{2 \cdot \text{mid}_{i,d}}$$

where $\text{mid}_{i,d}$ is the bid-ask midpoint. The monthly spread for each stock is the median of its daily half-spreads within the month, which is robust to days with stale or missing bid/ask quotes. Daily observations with a proportional half-spread above 2% are excluded as likely erroneous CRSP entries, following the convention in Novy-Marx and Velikov [19]. We apply this cost to BAB portfolio turnover: each rebalancing incurs one half-spread on the sold leg and one on the bought leg. Following Frazzini and Pedersen [13], we assume full monthly rebalancing, so this is an upper bound on realised costs.

Because the BAB factor scales each portfolio leg by the inverse of its beta, the transaction cost on each leg must be scaled accordingly. The round-trip cost borne per unit of BAB capital from the long leg is:

$$TC^L = \frac{2\bar{c}^L}{\beta^L}, \qquad TC^H = \frac{2\bar{c}^H}{\beta^H}$$

where $\bar{c}^L = \sum_i w_i^L c_i$ and $\bar{c}^H = \sum_i w_i^H c_i$ are the portfolio-weighted average monthly half-spreads of the low- and high-beta legs, respectively, and the factor of two reflects the full round-trip (entry and exit). The total net BAB return is then $r^{BAB,\text{net}} = r^{BAB} - TC^L - TC^H$.

Because reliable CRSP bid-ask data are only available from January 1993 onward, the TC analysis covers a shorter period than the main BAB analysis (January 1963 to December 2025). The gross BAB return over the TC subsample is 9.56% per year, compared to 11.00% over the full sample, so the two figures are not directly comparable.

Chapter 4

Results

In this chapter, we present the main results of our analysis. First, we showcase the prevalence of the beta anomaly by testing Proposition 1, then we present the performance of the BAB portfolio to test Proposition 2. Lastly, we present the results of our extension of Proposition 2 by introducing transaction costs into our analysis.

4.1 Proposition 1: The Security Market Line

To test Proposition 1 we run Fama-MacBeth cross-sectional regressions of monthly excess returns on lagged betas. The results are presented in Table 4.1. The intercept is positive and highly statistically significant with a mean of 1.17% per month and a t-statistic of 6.60, indicating that there is a substantial return that is unrelated to market beta. More central to proposition 1, the slope coefficient on beta is -0.19% per month with a t-statistic of only -0.71, which is statistically indistinguishable from zero. Under the CAPM, this slope should equal the market excess return, which over our sample averages considerably higher. The near-zero and insignificant beta coefficient therefore indicates that the cross-sectional compensation for bearing systematic risk is far smaller than the CAPM predicts, consistent with a flatter-than-predicted security market line and with the beta anomaly. We note that the Fama-MacBeth slope of -0.19% per month is only marginally negative, whereas the empirical SML plotted below appears to slope more clearly downward. This difference in magnitude reflects different weighting: Fama-MacBeth weights all individual stocks equally within each month, so the large mass of moderate-beta stocks that constitute the bulk of the cross-section anchors the average slope close to zero. The decile-based SML, by contrast, fits a line through ten equally-weighted group averages, giving the extreme high-beta decile (average beta 1.6) disproportionate leverage on the fitted slope and pulling it further negative. Both results are fully consistent with the beta anomaly and Proposition 1: the empirical compensation for beta is economically negligible and statistically indistinguishable from zero, far below the CAPM prediction.

Table 4.1: *Fama-MacBeth regression results. Time-series averages of monthly cross-sectional coefficients of excess returns on lagged beta. The mean is expressed in percent per month. t-statistics use Newey-West standard errors with four lags. Sample: US equities, January 1963 to December 2025.*

	Mean (%/month)	t-statistic
Intercept	1.17	6.60
Beta	-0.19	-0.71

4.1.1 Beta-Sorted Portfolio Alphas

The Fama-MacBeth results are complemented by an analysis of CAPM alphas across beta-sorted decile portfolios. Sorting all stocks into ten groups by their estimated beta and computing each decile's average monthly CAPM alpha. This shows that low-beta deciles earn positive and statistically significant alphas: decile 1 (average beta of 0.55) earns 0.24% per month, while deciles 2 and 3 earn 0.31% and 0.25% per month, both significant at conventional levels with t-statistics of 3.43 and 3.16, respectively. Moving up the beta distribution, alphas decline steadily. Decile 6 is essentially zero (0.06%, t=0.83), and the three highest-beta deciles earn negative and increasingly significant alphas. Decile 9 (average beta 1.37) earns -0.29% per month (t=-2.82), and the top decile (average beta 1.64) earns -0.50% per month (t=-3.16). This monotone decline in alpha across the beta distribution is precisely what proposition 1 predicts. Table 4.2 and Figure 4.1 and Figure 4.2 visualise these results: Figure 4.1 plots the empirical security market line (solid blue) against the CAPM prediction (red dashed), showing that the empirical line is clearly flatter; Figure 4.2 displays the CAPM alpha for each decile, illustrating the transition from positive alphas at the low end to large negative alphas at the high end.

Table 4.2: *CAPM alpha by beta decile. Value-weighted CAPM alpha for each beta-decile portfolio, estimated from time-series regressions of portfolio excess returns on the market excess return. Deciles are formed monthly by sorting on lagged Vasicek-shrunk beta. Green shading indicates positive alpha; red shading indicates negative alpha. Sample: US equities, January 1963 to December 2025.*

Decile	Avg. $\hat{\beta}$	Monthly α (%)	t-statistic
1	0.55	+0.24	2.14
2	0.68	+0.31	3.43
3	0.78	+0.25	3.16
4	0.86	+0.20	2.62
5	0.95	+0.13	1.79
6	1.03	+0.06	0.83
7	1.11	+0.02	0.27
8	1.22	-0.10	-1.28
9	1.37	-0.29	-2.82
10	1.64	-0.50	-3.16

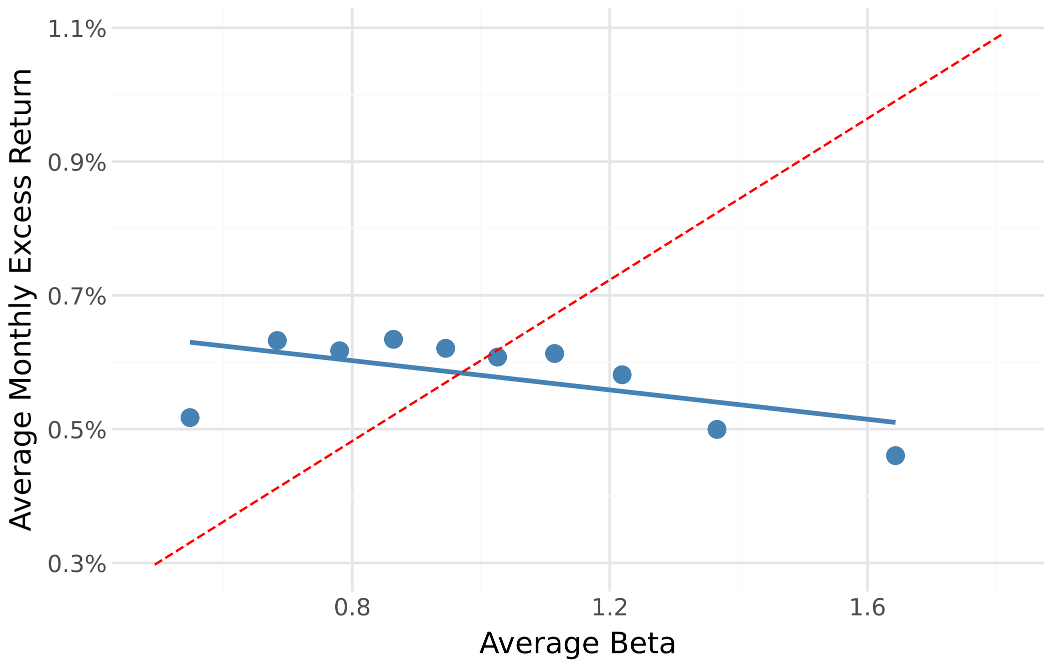

Figure 4.1: Empirical security market line (solid blue) estimated through decile-average betas and returns, against the CAPM prediction (red dashed). The flatter slope of the empirical line is consistent with the beta anomaly.

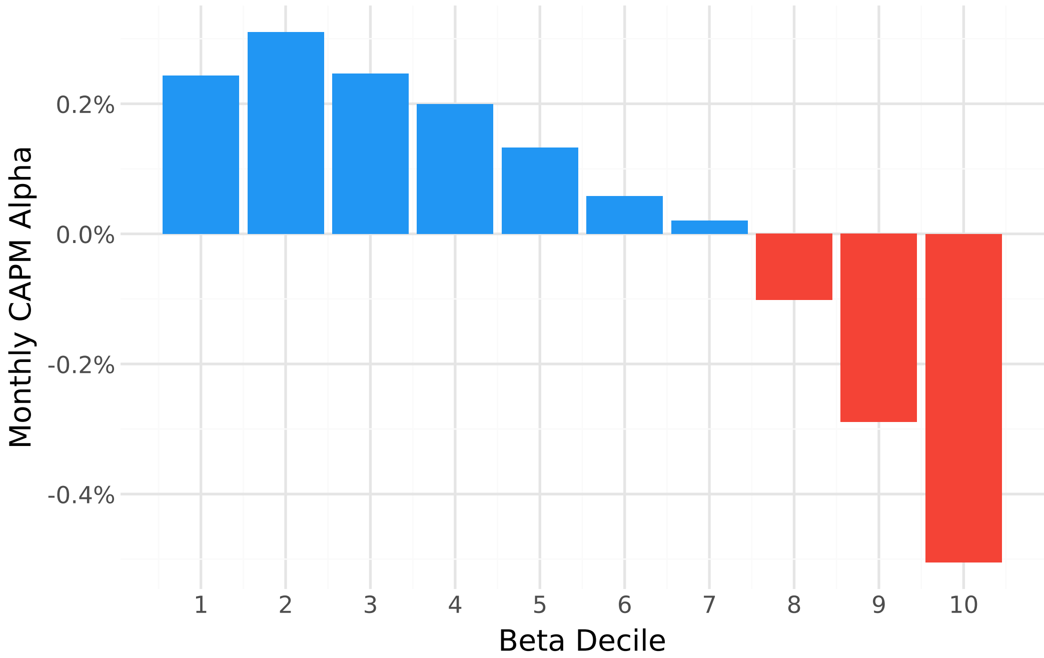

Figure 4.2: Monthly CAPM alpha by beta decile. Blue bars indicate positive alphas (low-beta deciles); red bars indicate negative alphas (high-beta deciles). The monotone decline from left to right is consistent with proposition 1.

4.2 Proposition 2: The BAB Factor

Having established the presence of the beta anomaly, we now turn to Proposition 2 and examine whether a BAB portfolio that exploits this anomaly generates positive excess returns. Following Frazzini and Pedersen [13], we construct the BAB factor each month by going long a leveraged portfolio of below-median-beta stocks and short a de-leveraged portfolio of above-median-beta stocks, with both legs scaled to a beta of one. The performance statistics are reported in Table 4.3. The BAB factor earns an annualised mean return of 11.00% with an annualised standard deviation of 11.61%, yielding a Sharpe ratio of 0.95. The associated t-statistic of 7.34 indicates that the positive mean return is highly statistically significant, providing strong support for proposition 2. Figure 4.3 illustrates the time-series behaviour of the strategy: starting from $1 in the early 1960s, the portfolio grows substantially through the 1990s, partially retraces during 2000–2010, before recovering and compounding further toward the end of the sample. The strategy thus generates meaningful returns over the full sample period, albeit with considerable variation across sub-periods.

Table 4.3: *BAB factor performance. Mean return and standard deviation are annualised by multiplying monthly values by 12 and $\sqrt{12}$, respectively. The Sharpe ratio is the ratio of annualised mean to annualised standard deviation. The t-statistic tests whether the mean monthly return differs from zero. Sample: US equities, January 1963 to December 2025.*

	BAB Factor
Ann. Mean Return	0.1100
Ann. Std Dev	0.1161
Sharpe Ratio	0.9473
t-statistic	7.3380

The BAB factor is constructed to be market-neutral, but estimation error in betas means this only holds in expectation. As an empirical check, we regress the monthly BAB return on the market excess return. The realised market beta is -0.046 (t = -1.66), confirming that the strategy carries negligible systematic market exposure. The annualised CAPM alpha is 11.32% (t = 7.50), closely matching the mean return in Table 4.3 as expected when market beta is near zero.

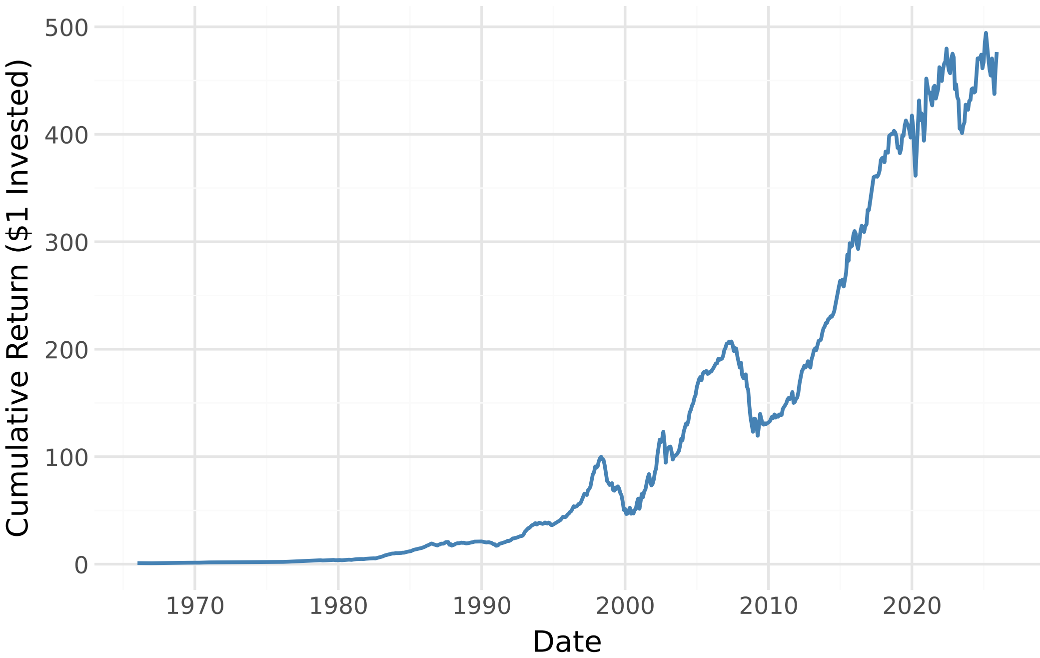

Figure 4.3: Cumulative return of the BAB factor, assuming $1 invested at inception. The strategy grows substantially through the late 1990s, retraces during 2000-2010, and partially recovers thereafter.

The strong cumulative returns in Figure 4.3 are achieved with embedded leverage. The long leg scales returns by $1/\beta^L$, and since the low-beta portfolio has $\beta^L < 1$, the long side invests on margin. Over our sample the average leverage multiple on the long leg is 1.53 — meaning every $1 of BAB capital controls roughly $1.53 of low-beta exposure — while the short leg is de-levered to a scaling factor of 0.71. The impressive compounding visible in the figure therefore partly reflects this structural leverage, and returns would be materially lower if the long leg were held unlevered.

Figure 4.4 decomposes the cumulative return into its long and short leg contributions separately, clarifying which side drives performance and when.

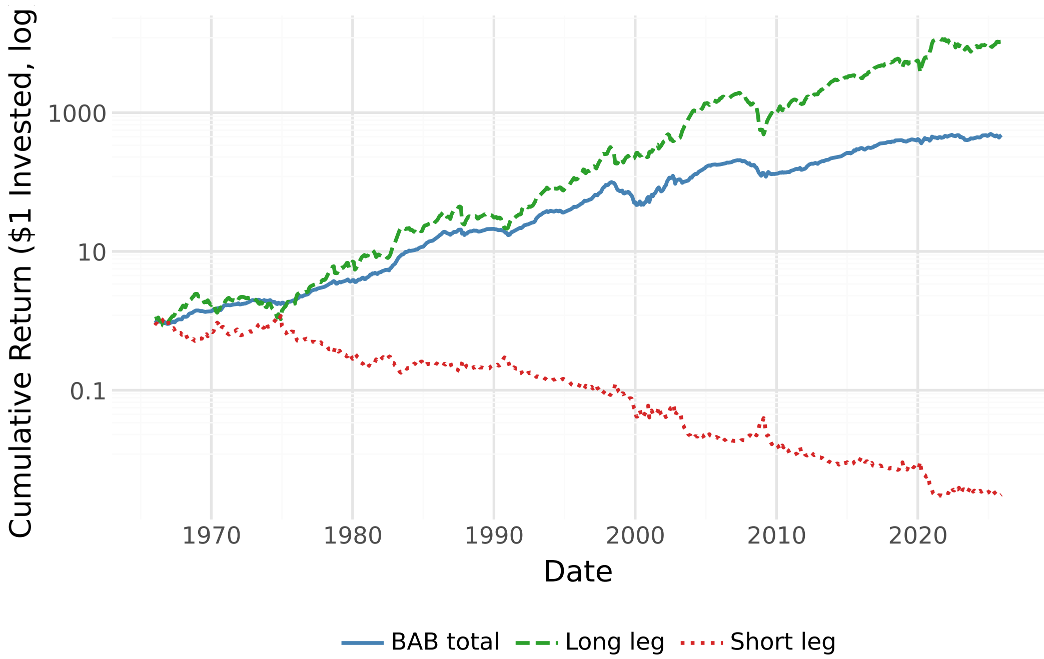

Figure 4.4: Cumulative return decomposition of the BAB factor by leg. The long leg compounds $(1 + r_e^L/\beta^L)$ and the short leg compounds $(1 - r_e^H/\beta^H)$, each starting from \$1. A rising short leg indicates the de-levered short position in high-beta stocks contributed positively. The BAB total combines both. Sample: US equities, January 1963 to December 2025.

Table 4.4 decomposes BAB performance across four sub-periods to examine when the strategy performs and when it falters. The periods reflect structural transitions in the data: the pre-1990 early period; the strong 1990s bull market; the 2001–2009 decade encompassing the dot-com crash and the global financial crisis; and the post-crisis period from 2010 onward.

Table 4.4: *BAB factor performance by sub-period. Ann.~Mean and Ann.~Std are annualised (multiplied by 12 and $\sqrt{12}$). Sharpe is the ratio of annualised mean to annualised standard deviation. The t-statistic tests whether the mean monthly return differs from zero. N is the number of months in each sub-period.*

	Ann. Mean	Ann. Std	Sharpe	t-stat	N
1963–1989	0.1324	0.0925	1.4305	7.0082	288.0000
1990–2000	0.1055	0.1362	0.7741	2.5673	132.0000
2001–2009	0.1009	0.1773	0.5689	1.7067	108.0000
2010–2025	0.0846	0.0860	0.9834	3.9335	192.0000

4.3 Transaction Costs

While the gross BAB factor earns positive returns, a real-world investor must also bear the transaction costs of monthly rebalancing. To assess this, we estimate round-trip costs from daily CRSP bid-ask spreads and apply them to the turnover of each portfolio leg. The results, reported in Table 4.5, reveal a dramatic impact of trading frictions. Over the subsample for which transaction cost data are available, the gross BAB strategy earns an annualised mean return of 9.56% (t= 4.15), but after deducting estimated transaction costs the net return falls to -23.51% per year (t= -9.29). The average annual transaction cost drag amounts to 33.07 percentage points, far exceeding the gross return. This implies that, under the assumption of full monthly rebalancing, the BAB strategy is entirely non-viable under these upper-bound cost assumptions. The result should be interpreted as an upper bound on actual transaction costs, since in practice investors would rebalance less frequently and only trade when the cost savings exceed the friction. Notably, the estimated annual TC drag of 33.07 percentage points exceeds even the full-sample gross BAB return of 11.00%, meaning the strategy would be non-viable under these cost assumptions regardless of which sample period is used to measure gross returns. Nonetheless, the result highlights that the gross profitability documented in section 4.2 is highly sensitive to implementation costs, particularly given the wide bid-ask spreads prevalent in CRSP data for earlier decades prior to market decimalization.

Table 4.5: *Gross and net BAB returns over the transaction cost subsample (January 1993 to December 2025). Transaction costs are estimated from CRSP daily bid-ask half-spreads assuming full monthly rebalancing. Net BAB deducts the round-trip cost on both portfolio legs. Avg annual TC drag is the mean annual cost. t-statistics test whether the mean monthly return differs from zero.*

	Gross BAB	Net BAB (after TC)	Avg annual TC drag
Ann. mean return	0.0956	-0.2351	0.3307
t-statistic	4.1466	-9.2920	nan

The aggregate TC drag conceals substantial variation across market structure regimes. US equity markets transitioned from fractional to decimal tick sizes between June 2000 and April 2001 (NYSE and AMEX fully decimal by January 29, 2001; NASDAQ by April 9, 2001), compressing bid-ask spreads by roughly 80% on average. Table 4.6 decomposes the TC drag into three sub-periods aligned with this structural break: the pre-decimalization era (1993–2000), the post-decimalization decade that overlaps with Frazzini and Pedersen's (2014) original sample end (2001–2012), and the modern era (2013–2025).

The decomposition reveals that the aggregate drag is overwhelmingly a pre-decimalization phenomenon. During 1993–2000, the TC drag reaches 68.8 percentage points per year, driven by average monthly long-leg half-spreads of 1.18% — consistent with the 12.5 cent minimum tick on fractional markets, which translates to several percent for the small-cap, low-beta stocks that anchor the long leg. The leverage amplification built into the BAB formula ($TC^L = 2\bar{c}^L/\beta^L$) then multiplies this raw spread cost by roughly $1/\beta^L \approx 2$ to 3. After decimalization, the drag falls sharply: to 28.8 pp/year in 2001–2012 and 15.0 pp/year in 2013–2025, with corresponding net BAB returns of -18.4% and -7.3% per year, respectively. These findings materially qualify the conclusion from Table 4.5: the estimated TC drag declines substantially after decimalization, and the aggregate non-viability documented in the full TC subsample is driven overwhelmingly by the pre-decimalization era. Even so, the estimated net returns remain negative in all three sub-periods. Whether the strategy is viable under modern market conditions therefore depends on the degree to which our quoted-spread estimates overstate actual execution costs for institutional investors — an interpretation taken up in Section 5.3.

Table 4.6: *TC drag decomposition by market structure era. Gross BAB, TC Drag, and Net BAB are annualised (% per year). Spread^L is the average monthly half-spread on the long leg (% per month), before leverage scaling. t (Net) tests whether the mean monthly net return differs from zero. The decimalization break (January 2001) separates the first two periods.*

	Gross BAB	TC Drag	Net BAB	t-stat	Spread^L	N
1993–2000	11.18	68.79	-57.62	-10.87	1.18	96.00
2001–2012	10.42	28.80	-18.38	-3.99	0.59	144.00
2013–2025	7.78	15.03	-7.25	-2.89	0.38	156.00

Chapter 5

Discussion

In this chapter we discuss the implications of our findings, compare them to the existing literature on the beta anomaly, and address the limitations of the analysis.

5.1 Proposition 1

Our finding that the empirical security market line is significantly flatter than the CAPM predicts is consistent with empirical evidence. The seminal study of Black, Jensen, and Scholes [6] was among the first to document this phenomenon systematically, showing through portfolio time-series tests on NYSE stocks from 1926 to 1966 that the relationship between beta and average returns is too flat to be reconciled with the CAPM. Subsequent work by Fama and French [9] confirmed this result using cross-sectional regressions and further showed that once size and book-to-market are included, beta loses essentially all explanatory power for the cross-section of expected returns. Fama and French [10] revisit the evidence and conclude that the CAPM's core prediction — a proportional, positive relationship between beta and expected return — is contradicted by the data.

Our Fama-MacBeth results are broadly consistent with these findings. The estimated beta coefficient of -0.19% per month is a small fraction of the average market excess return over our sample, and it is not statistically distinguishable from zero. The decile alpha analysis further reinforces this: the alpha spread from the lowest to the highest beta decile is approximately -0.75 percentage points per month, a magnitude that is both economically large and statistically significant at the top and bottom of the distribution.

Frazzini and Pedersen [13] document the same qualitative pattern across US equities from 1926 to 2012 and 20 international markets, providing the theoretical framework through which funding constraints and leverage limits generate precisely the flat or inverted alpha-beta relationship we observe. Our results replicate their core finding for the US equity market over a partly overlapping and partly extended sample period (1963–2025), lending further out-of-sample support to the robustness of proposition 1. The flat SML is also related to the broader low-risk anomaly documented internationally by Blitz and van Vliet [7], and to the finding of Ang et al. [2] that stocks with high idiosyncratic volatility earn anomalously low returns — suggesting that the mispricing of risk is not limited to systematic beta but extends across measures of total and idiosyncratic risk.

5.2 Proposition 2

Our BAB factor earns an annualised mean return of 11.00% with a Sharpe ratio of 0.95 and a t-statistic of 7.34, which we interpret as strong support for proposition 2. The magnitude is broadly in line with what Frazzini and Pedersen [13] report across their US sample, suggesting the anomaly has remained robust over our extended sample period through 2025. A caveat when interpreting the level of returns and the cumulative chart is that the strategy does not operate at zero leverage: the long leg is scaled by $1/\beta^L$, where $\beta^L < 1$, imposing a leverage multiple that over our sample averages approximately 1.53. Returns from an unlevered long-only position in low-beta stocks would therefore be substantially lower than those reported for the BAB factor.

Baker, Bradley, and Wurgler [4] argue that the low-beta anomaly creates a structural tension between benchmark-constrained investors — who mechanically demand high-beta stocks to keep up with their benchmark and unconstrained arbitrageurs who would otherwise eliminate the mispricing. As the BAB strategy became widely known following its formalisation by Frazzini and Pedersen [13], increased capital allocated to exploiting it may have moderated returns over time. A formal test of post-publication decay would require splitting the sample around 2014, which lies outside the scope of our analysis. Nonetheless, the gross BAB return over the 1993–2025 subsample is 9.56% per year (t = 4.15), broadly in line with the 11.00% (t = 7.34) earned over the full 1963–2025 sample, offering tentative evidence that the anomaly has not disappeared in the more recent period.

5.3 Transaction Costs

The most striking result of our analysis is the magnitude of the estimated transaction cost drag: 33.07 percentage points per year. This figure should be treated as an upper bound that almost certainly overstates the true costs an investor would face, for reasons detailed below, but it is nonetheless informative in showing that gross BAB profitability is not immune to frictions. Under our baseline assumptions it completely overwhelms the gross return, rendering the net strategy loss-making.

Several factors inflate our estimate. First, the CRSP quoted bid-ask spreads are particularly wide in the pre-decimalization era. Prior to the phased introduction of decimal pricing on US exchanges in 2001, minimum tick sizes were as large as one-eighth of a dollar, mechanically inflating quoted spreads. A significant portion of our transaction cost subsample (which begins in 1993) thus falls in a period of structurally high spreads that are not representative of the execution environment available to large investors. Second, we assume full monthly rebalancing of all positions, which is an upper bound on turnover. In practice, an investor would apply turnover filters and tolerate small deviations from target weights, substantially reducing costs. Third, quoted spreads systematically overstate effective spreads: institutional investors routinely execute inside the quote, and effective spreads are typically closer to half of quoted spreads for liquid stocks.

These concerns are directly addressed by Frazzini, Israel, and Moskowitz [12], who use proprietary data from an actual large asset manager to measure the trading costs of factor strategies including BAB. Their analysis finds that effective implementation costs for such strategies are on the order of tens of basis points per year, far below our estimate, and that the BAB strategy remains profitable after these realistic costs. Importantly, Frazzini, Israel, and Moskowitz [12] also report full round-trip costs (covering both entry and exit), as we do — the factor of two applied to the half-spread in our methodology is therefore not a source of the discrepancy. The divergence instead stems entirely from the three sources described above: quoted versus effective spreads, pre-decimalization tick inflation, and the full-turnover assumption. Our result is therefore best understood as a sensitivity analysis that sets an extreme upper bound, demonstrating that the gross profitability is not infinitely robust to frictions, without implying that the strategy is inherently unprofitable after realistic implementation. This conclusion is consistent with the broader findings of Novy-Marx and Velikov [19], who systematically evaluate transaction costs across a large set of published anomalies and find that while many strategies are substantially eroded by costs, the most liquid and well-known factors tend to survive after realistic friction adjustments.

5.4 Limitations and Further Research

Our analysis has several limitations that suggest directions for future research. First, our risk adjustment relies solely on the CAPM. The BAB alpha could shrink under richer factor models such as the Carhart [8] four-factor model or the Fama-French [11] five-factor model, which control for size, value, momentum, profitability, and investment. Frazzini and Pedersen [13] show that the BAB alpha survives these adjustments in their original sample, but verifying this holds over our extended sample through 2025 would strengthen the robustness of the conclusions.

Second, our study is restricted to US equity markets, while Frazzini and Pedersen [13] demonstrate that the beta anomaly and BAB profitability are pervasive across 20 international equity markets as well as Treasury bonds, corporate bonds, and futures. Examining whether our specific sample-period results generalise internationally would provide a more complete picture of the anomaly's robustness.

Third, our transaction cost analysis relies entirely on CRSP bid-ask quotes, which suffer from the limitations described above. A more rigorous approach would follow Frazzini, Israel, and Moskowitz [12] in distinguishing between the pre- and post-decimalization periods, applying effective-spread estimates rather than quoted spreads, and incorporating realistic turnover constraints. Restricting the cost analysis to post-2001 data would provide a more realistic picture of contemporary implementation costs.

Fourth, while CRSP is generally regarded as free of survivorship bias for US common stocks, our beta-sorted portfolios may be subject to delisting return bias. High-beta stocks are more likely to be small, volatile, and financially distressed, and their delisting returns are often negative and difficult to measure accurately. If these missing or mismeasured delisting returns are disproportionately concentrated in the high-beta deciles, the alpha estimates there may be upward biased. A robustness check applying explicit delisting return corrections would address this concern.

Finally, our analysis holds fixed the 1963–2025 sample as a single period. Given the structural break introduced by decimalization and the growing institutional awareness of the beta anomaly in the post-2000 period, a sub-period analysis comparing pre- and post-2001 results would illuminate whether the anomaly and the BAB strategy have weakened over time as the literature would predict.

Chapter 6

Conclusion

This paper contributes to the literature on market anomalies by testing Propositions 1 and 2 from Frazzini and Pedersen (2014) [13], which derive testable predictions about the beta anomaly and the performance of the BAB strategy. Using data from US equity markets we find evidence of a negative relationship between beta and alpha, supporting Proposition 1 and confirming that the empirical Security Market Line (SML) is significantly flatter than the Capital Asset Pricing Model (CAPM) predicts. We also find that the BAB portfolio generates positive risk-adjusted returns (alpha), supporting Proposition 2. However, when we introduce transaction costs under the assumption of full monthly rebalancing using CRSP quoted spreads, the estimated cost drag overwhelms the gross return, rendering the net strategy loss-making — though this estimate almost certainly represents an upper bound on true implementation costs, as argued by Frazzini, Israel, and Moskowitz [12].

The limitations of this paper suggest several directions for future research. Most directly, our risk adjustment relies solely on the CAPM, and BAB alpha could shrink under richer factor models such as those of Carhart [8] and Fama and French [11]; verifying robustness to these adjustments would strengthen the conclusions. Additionally, the analysis is restricted to US equity markets, which may limit generalisability, and the transaction cost estimates rely on quoted rather than effective spreads, likely overstating true implementation costs. Sub-period analyses splitting around the paper's 2014 publication date would further illuminate whether the anomaly has weakened as factor investing has grown.

Taken together, our results suggest that the beta anomaly is a robust feature of US equity markets over the past six decades, and that the structural forces identified by Frazzini and Pedersen — leverage constraints and benchmark-driven demand for high-beta stocks — remain relevant explanations. Whether the anomaly can be profitably exploited in practice depends critically on transaction costs, a question that warrants continued attention as data on actual trading costs become more widely available.

Chapter 7

Appendix

7.1 Code Availability

The full replication code for this paper is available at the following GitHub repository:
https://github.com/gfs189/Seminar-Asset-Prices-and-Financial-Markets

Chapter 8

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